

WENTWORTH LIU

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Data Scientist | Machine Learning Engineer | Applied ML | Statistical Modelling & Decision Science

SUMMARY

Data Scientist and Machine Learning Engineer with 2+ years of experience turning complex EV-fleet data into causal decision systems, interpretable ML workflows, reusable feature infrastructure, and AI-enabled internal products. Combines first-principles materials-science training with Python, SQL, PySpark, statistical modelling, MLOps, and cross-functional product delivery across battery safety, R&D, and after-sales operations.

EXPERIENCE

Li Auto Battery Intelligence Department
Data Scientist Jul 2024 – Present

- **Causal Decision Science for Battery Longevity & Lifecycle Strategy.** Led a business-led causal study on million-scale fleet telemetry to identify actionable longevity levers amid observational confounding. Applied propensity-score matching, covariate-balance diagnostics, and ATT estimation to separate likely causal effects from confounders and intermediate variables. Converted the evidence into self-service monitoring and decision workflows; partnered with BMS, cell-design, performance, and after-sales teams on feasible usage/OTA strategies, design-boundary validation, and a scenario-based single-model warranty-saving projection in the tens of millions of RMB.
- **Rare-event Safety Risk Scoring & ML Operations.** Addressed thermal-runaway risk where scarce failure labels and diagnostic needs made a conventional binary classifier unsuitable. Reframed the problem as explainable relative-risk ranking, then built a monthly ML workflow spanning data-quality gates, platform-aware feature normalisation, statistical feature selection and redundancy control, evidence-weighted scoring, contribution-level explanations, batch scoring, publication, and audit checks. Deployed across five battery platforms on the internal ML platform, capturing 93% of newly observed failures within the top 2% of risk-ranked vehicles; enabled targeted inspection, case review, and root-cause investigation with after-sales, quality, and R&D.
- **Feature Store & Reusable ML Data Platform.** Designed and built a department-wide offline Feature Store to replace fragmented notebooks with a shared, configuration-driven system used and co-maintained by the team. Created a Git-controlled YAML feature registry and metadata/lineage model; implemented PySpark ODS-to-Hive-DM transformations covering event semantics, temporal alignment, start-end joins, time-weighted and cross-signal aggregation, platform-schema mapping, and selective materialisation. Standardised reusable features and train-score-diagnose consistency across five platforms, reducing feature-development cycle time by ~85%.
- **AI-enabled Risk Intelligence Product & System Architecture.** Designed and AI-assisted-developed a cloud-deployed internal workbench that turns model and fleet-data outputs into feature visualisation and multi-perspective high-risk investigation. Defined a maintainable Web/API/asynchronous-worker architecture with StarRocks-backed retrieval, IDaaS-based access control, and auditable replay; integrated cohort comparison, single-vehicle analysis, operating-condition distributions, similarity analysis, and historical TR evidence into one workflow. The product is actively used internally and is expanding to additional modules.
- **Human-in-the-Loop NLP Quality Intelligence.** Co-designed a 30+ category taxonomy with after-sales experts and built a hybrid rules + TF-IDF + Random Forest classifier for 15k+ noisy work orders. Established confidence-based routing, expert-term evaluation, error analysis, and closed-loop retraining; improved operational accuracy from 85% to 96% within three months, achieved Weighted F1 > 0.90, and replaced roughly 12 hours of weekly manual triage.

ByteDance Douyin Market Strategy
Data Analytics Intern May – Sep 2023

- **Multi-source Intelligence & Growth Decision Analytics.** Integrated signals from 5+ data platforms into an IP-popularity scoring system, added monitoring and workflow automation, and redesigned content metrics and classification frameworks to guide campaign and resource-allocation decisions; drove ~60% dashboard adoption and validated ~40% consumption growth following targeted campaigns.

Dongwu Securities Mechanical Sector
Equity Research Intern Dec 2021 – Mar 2022

- **Quantitative Market Research & Forecasting.** Analysed fundamentals, competitive positioning, and market dynamics for Harmonic Drive Systems; built a stock-price forecasting model and delivered investment recommendations.

EDUCATION

Nanjing University Sep 2021 – Jun 2024
MSc Computational Materials Science · Developed *Skeaf*, a 3D Fermi-surface visualisation tool combining HDBSCAN clustering with quantum-oscillation data; first author, *Physical Review B* (2024); co-author, *Physical Review B* (2024) and *Physical Review Materials* (2023); Patent: CN115795266A.

Nanjing University Sep 2017 – Jul 2021
BSc Materials Science & Engineering (Outstanding Graduate) · GPA: 4.6/5.0 · Top 1% of cohort · National Scholarship (top 0.15%) · MCM/ICM 2020 Honorable Mention (Team Captain) · Sakura Science Exchange Programme, Tohoku University.

SKILLS

Programming	Python · SQL/HiveSQL · PySpark · C++ · MATLAB
Causal & Applied ML	PSM · ATT · Covariate Balance · Statistical Modelling · Rare-event ML · Feature Normalisation · Risk Ranking · NLP · Random Forest
ML Engineering	Feature Store Design · Feature Registry · Metadata & Lineage · Batch ML Pipelines · Data-quality Gates · Model Auditability · Internal ML Platform Deployment
Systems & Product	Analytical Workflow Design · Product Design · Web/API/Worker Architecture · StarRocks · Asynchronous Analysis · IDaaS · Data Productisation
AI Delivery	Agent Workflow Design · LLM Integration · Prompt Engineering · Codex · Claude Code
Domain & Lang.	Battery Intelligence · EV Analytics · Engineering Systems · English (TOEFL 104) · Mandarin/Cantonese (Native) · Japanese (JLPT N1)